

MARC
Management

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1.0 Introduction

With the advancement of Technology in today's Electrical & Hydraulic Systems, it is a requirement that Service Technicians have ready access to a readable System Schematic. It is important for these schematics to be readily accessible for Shop and Field use, as well as being large enough to read, durable enough to withstand wind & rain, and allow for temporary marking with grease pencils during the trouble-shooting process.

2.0 Best Practice Description

Newmont's Phoenix Mine has created a cabinet that stores laminated copies of hydraulic and electrical schematics for all their mobile equipment. This cabinet is located in the Main Shop and is easily accessible to all technicians. Technicians can remove a laminated schematic, take it to the machine they are working on, mark it up with a grease pencil as they trace wires or hoses and work their way through a diagnostic fault-tree.

When they are finished with the schematic, it can be easily wiped clean and placed back in the cabinet for the next use.

3.0 Implementation Steps

Cabinet

Deciding what type of Cabinet is available or that space constraints allow for.

Ideally the cabinet should allow the schematics to hang in a way they can dry if they are wet.

Newmont's cabinet is 18" deep x 39" wide x 54" height.

By having two rods in one direction and two in the other direction, you can quickly sort through and find the schematic you want and then slide all to the rearward area, the others to the front, then when you open the front panel, the one you want is easily removed from the rods.

Schematics

"Blown Up" schematics should be printed and laminated in the appropriate size to hang in the selected cabinet. The schematics are taken directly from the CAT service manual.

The schematics should hang in a way that when the cabinet is opened, the description of each schematic is readily visible.

On the back side of each schematic, critical system spec's are provided for reference.

At least once a year the schematics should be checked to verify they still match what is available in SIS.

4.0 Benefits

This resource facilitates a more rapid and accurate trouble-shooting & diagnostic process because technicians are inclined to fully utilize these accessible, user friendly schematics.

These schematics are relatively inexpensive to produce, so they can be easily updated any time SIS changes are made to the source document.

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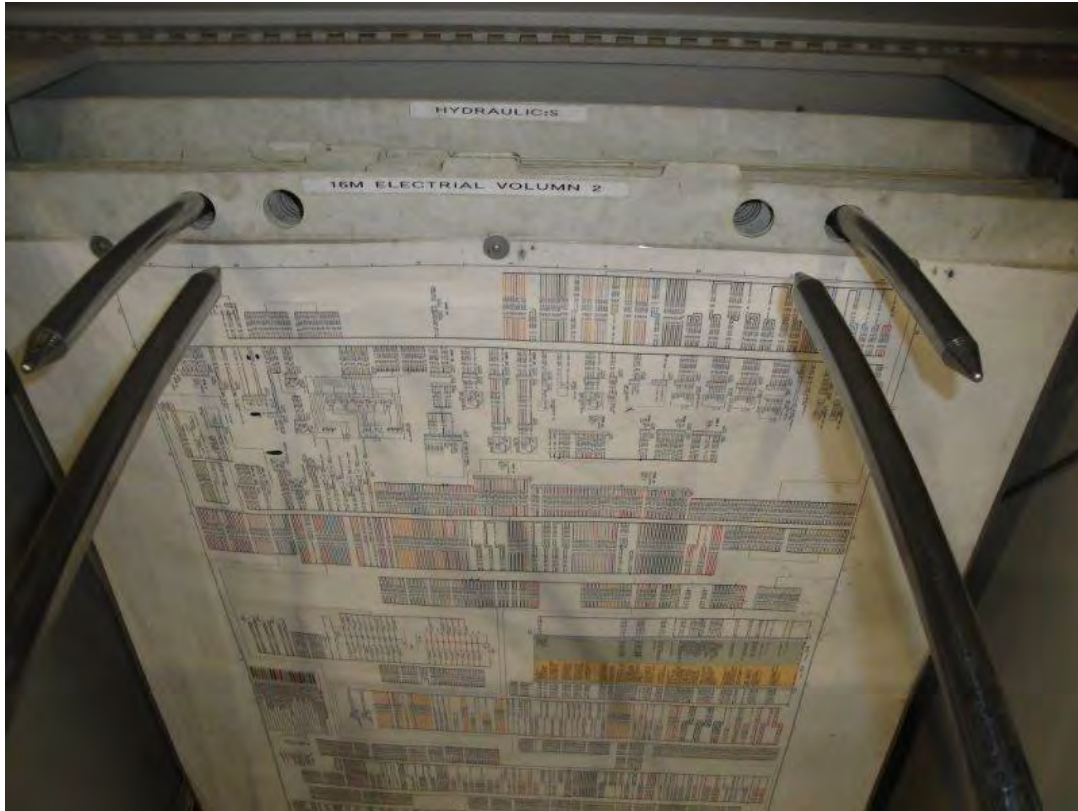
5.0 Resources Required

- Cabinet
 - o The file cabinet in this example is made by:
 - o Easi File Corporation
 - o Irvine, California. Ph 1-714-855-4121
 - o Easi File Model no. EF200
- Laminated Schematics

6.0 Supporting Attachments / References



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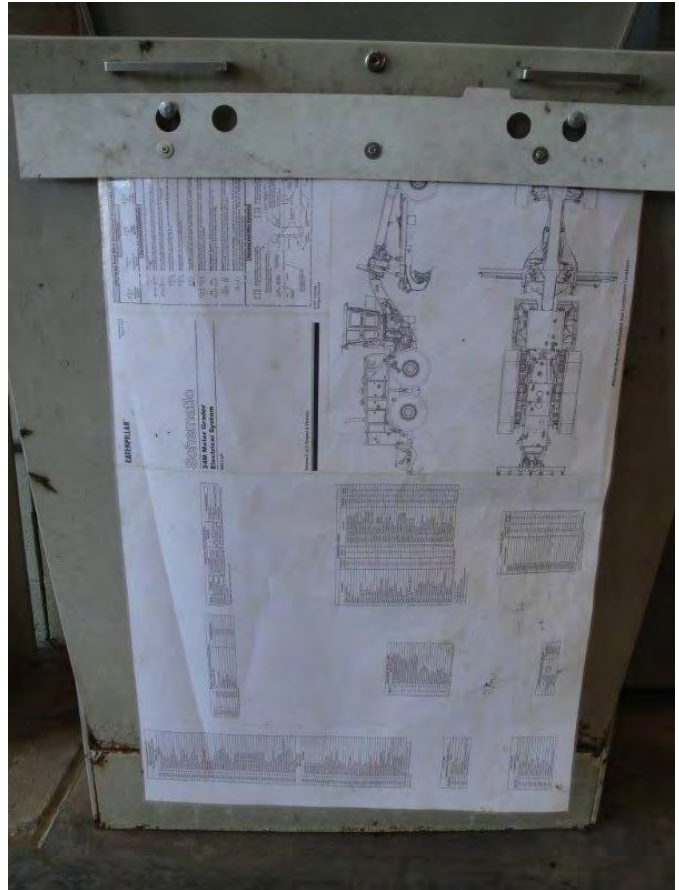
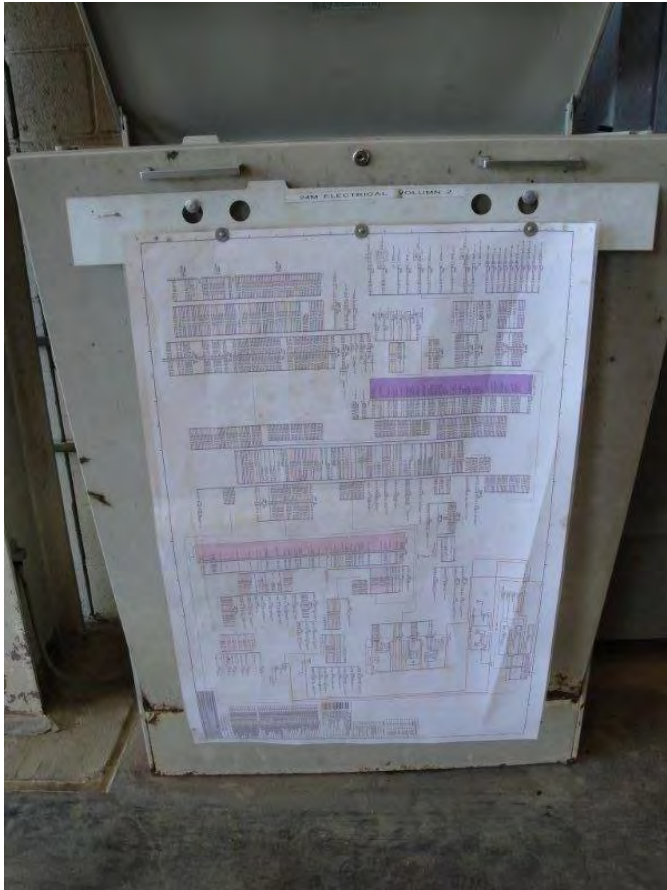
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7.0 Related Best Practices

8.0 Acknowledgements

Newmont's Phoenix Site Tige Brown
 Cashman's Phoenix Site Manager Phil Kendall
 Caterpillar's Product Performance Manager Howard Woolard

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